

Forecasting yield in temperate fruit trees from winter chill accumulation

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Abstract

The potential yields of deciduous fruit and nut trees are strongly affected by the climatic conditions that regulate dormancy and subsequent flowering. However, the realized yield at the end of the growing season depends on several additional factors including environmental conditions and management measures. These modulating factors dilute the direct effect of chill accumulation and frustrate efforts to forecast yields. Such forecasts are also limited by a lack of data, which complicates the assessment of the relationship between chill and yield. However, information on how diminishing chill impacts crop yield is crucial for farmers and other decision-makers, particularly in warm fruit and nut growing regions. We address this challenge by adopting a probabilistic approach to yield forecasting. This approach incorporates uncertainty and generates predictions that do not consist of precise numbers, but yield expectations expressed as probability distributions. We demonstrate a set of functions that we developed in the R programming language to generate forecasts of possible yields with given chill. We apply these methods to data sets of two sweet cherry (*Prunus avium* L.) cultivars 'Lapins' and 'Brooks'.

Keywords: flowering, dormancy, climate change, uncertainty, ripening, nut trees

INTRODUCTION

The yields of temperate fruit and nut trees depend on a range of biological processes such as budburst, flowering, and leafing. Exposure to cultivar-specific periods of low temperatures (chill requirement – CR) during winter allows trees to overcome their dormant stage (Campoy et al., 2011; Luedeling, 2012). After fulfilling this need for chill, dormant buds are able to grow and develop flowers and leaves (Campoy et al., 2011). Dormancy release usually coincides with increasing temperatures in early spring.

Warm winters can lead to insufficient chill availability, which can affect both dormancy and dormancy release processes. Effects can include a decrease in the percentage of budburst, delay in flowering dates, heterogeneous and protracted bloom periods, and, in extreme cases, bud abortion (Erez, 2000; Lavee and May, 1997). Chill deficiencies are widely expected to impact yield in species such as almonds, for which the number of flowers that become fruits largely determines the final yield (Lampinen et al., 2011). In other species such as sweet cherry, synchrony in bloom timing between pollen donors and receptors is crucial for obtaining adequate fruit set and subsequent yields (Faust and Surányi, 1996). However, chill accumulation is only one among many factors that determine yield, with many processes that occur between budburst and harvest having a strong impact on final production outcomes.

Farmers can mitigate the initial negative effects of warm winters through various management measures, such as the application of plant growth regulators or thinning to increase the final size of each fruit. Even if trees receive sufficient chill accumulated in winter, inclement events such as pre-harvest rains can substantially decrease fruit or nut yields (Knoche and Winkler, 2017). Such events can obscure the actual influence of chill accumulation on final yields. For example, Benmoussa et al. (2017) found that even after

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relatively cold winters pistachios still had low yields in many years. Similarly, Pope et al. (2015) found that chill accumulation of 40 chill portions (CP; i.e. much greater than the assumed CR of around 23 CP) in ‘Nonpareil’ almond was associated with a decrease by 20% in potential yield, noting that studies on the relationship between chill and yield should include factors such as climate change as well as other physiological parameters. However, identifying and establishing relationships between all important variables would likely require long-term research and complex modeling approaches. In the short term, a probabilistic modeling approach for projecting yield based on accumulated chill may help farmers and other actors associated with fruit and nut industries, as well as policy-makers. Here, we demonstrate an approach to studying the relationship between chill accumulation and yield based on a kernel density estimation algorithm to better forecast yield outputs. We showcase the use of this approach by estimating future yields for an experimental sweet cherry orchard in Quillota, Chile.

MATERIALS AND METHODS

Accounting for uncertainty in yield projections

We generated and tested four main functions as a collection of procedures that embrace the inherent uncertainty involved in the relationship between chill and yield. We included all four functions in the ‘pasitR’ package (Schiffers et al., 2018), an open-access extension package for the R programming language:

1. In order to explore the data, we generated the ‘chillscatter’ function that visualizes the relationship between accumulated chill (in CP) and yield. The linear fit of yield as a function of chill is based on a loess-smoothing procedure from the ‘plyr’ package (Wickham, 2011). ‘chillscatter’ also plots the histograms for the observations and the Spearman correlation coefficient.
2. We created the ‘chillkernel’ function to estimate the relative probabilities of yields across all observed chill accumulation levels. This approach smooths the probability surface using a two-dimensional kernel, which makes it particularly useful for sparse and scattered data. At its core is the ‘kde2d’ function of the ‘MASS’ package (Venables and Ripley, 2002).
3. We created the ‘chillkernelslice’ function to visualize the probability distribution of yield given a specific level of chill accumulation (in CP), based on a slice of the two-dimensional probability surface estimated with ‘chillkernel’. Since this is a cut through the 2D probability density function, the probability values have to be interpreted as relative, not as absolute probabilities.
4. We generated the ‘chillkernelslicerange’ function, to account for the inability to know the exact chill accumulation (in CP). This is a particularly important issue for projections based on weather forecasts. To include this uncertainty, we binned chill accumulation levels using the Freedman-Diaconis rule (Freedman and Diaconis, 1981) for calculating optimal bin width and estimating the relative probabilities across yields for the range of chill portions defined by these bins.

Experimental site and data recording

Yield data was recorded in the experimental orchard of the Escuela de Agronomía at the Pontificia Universidad Católica de Valparaíso, in Quillota, northern-central Chile. These records contained seasonal observations of marketable production for two sweet cherry cultivars (‘Lapins’ and ‘Brooks’) for the period from 2010 to 2017. Trees were planted at a spacing of 4.5 m between rows and 2.5 m within rows, resulting in a tree density of 888 trees per ha for both cultivars. Weather data, recorded from a nearby weather station (32.89°S, 71.20°W, 130 m above sea level), was downloaded from the Center for Climate and Resilience Research ([CR]²) sponsored by the University of Chile (www.cr2.cl). We selected daily minimum and maximum observations from the 1st of January 2010 to the 31st of December 2017.

Winter chill was calculated with functions from the ‘chillR’ package (Luedeling, 2019).

We estimated chill in chill portions according to the Dynamic Model (Erez et al., 1990; Fishman et al., 1987a, b), developed to estimate rest completion in deciduous fruit trees. For every year, we defined the winter season as the period between the 1st of May and the 31st of August.

Analyses and figures

All analyses and figures were implemented in the R programming language (R Core Team, 2019). The functions we used here are available in the 'pasitR' package (Schiffers et al., 2018), an open-access extension package for the R programming language.

RESULTS AND DISCUSSION

The analysis of the relationship ('chillscatter' function) between chill accumulation and yield showed a low correlation in both the 'Lapins' and 'Brooks' cultivars (Figure 1), with Spearman correlation coefficients of 0.43 for 'Lapins' and 0.30 for 'Brooks'. These overall low coefficients in both cultivars may be caused by our considering only eight years of data, one out of which (i.e. year 2015) included two important rain events (i.e. 21.4 mm and 32.8 mm) during the fruit growing stage. Such events may have substantially reduced the yield, even in a year with adequate winter chill accumulation (between 1st May and 31st August) of 43 CP, which is usually considered sufficient for the local cherry cultivars. Earlier studies from other locations have estimated the CR for bud-break at 37 CP for 'Brooks' (Erez, 2000) and 35 CP for 'Lapins' (Albuquerque et al., 2008), and these figures appear to be roughly applicable in the Quillota region.

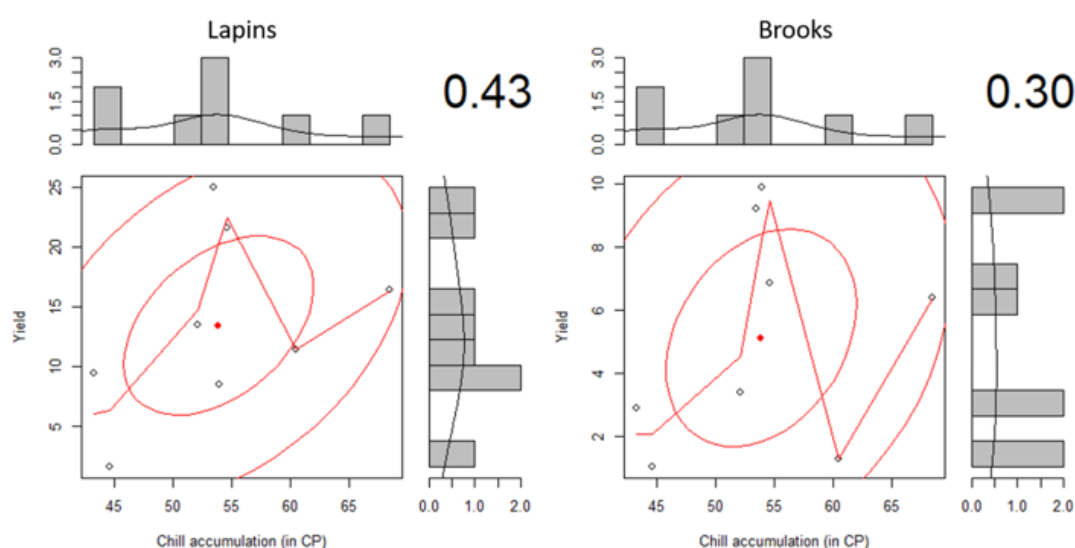


Figure 1. Scatter plot for chill accumulation (x-axis in chill portions – CP) and seasonal yield (y-axis in t ha⁻¹) for two sweet cherry cultivars ('Lapins' and 'Brooks') produced with the 'chillscatter' function. In each panel, white diamonds represent observations and red diamonds represent the mean. The red line across the plot shows the linear fit, red ellipses represent one and two standard deviation units from the mean. Grey bars on top and at the right side of the panel represent histograms for the observations. Black lines in each histogram plot represent the smoothed density curves. The number in the upper right corner represents the Spearman correlation coefficient for correlations between variables.

Along the same lines, frost events may also reduce the yield in some years if they occur during sensitive periods such as flowering. Our data set included one year (i.e. 2012) with a

long frost event (i.e. four hours below 0°C) with a minimum temperature of -1.6°C that occurred during the flowering period. According to Negron (2005), this temperature can lead to flower damage and fruit set problems, affecting the yield by reducing the total number of fruits. While the correlation of chill and yield was thus relatively low, it may have been similar to the relationship reported by Benmoussa et al. (2017) for the pistachio cultivar ‘Mateur’ in Tunisia (although they do not specifically report a correlation coefficient), which the authors reported as positive and statistically significant.

The ‘chillkernel’ analysis applied to our data set of eight years of records showed that chill accumulation between approximately 50 and 60 CP during the winter season resulted in high probabilities of achieving large yields in both ‘Lapins’ and ‘Brooks’ cultivars (Figure 2). Given the mentioned range of chill accumulation, most likely yields ranged between 10 and 25 tons ha⁻¹ for ‘Lapins’ and between 6 and 10 tons ha⁻¹ for ‘Brooks’. The optimal chill interval for this data set, estimated with the Interquartile Range, appeared to be 49.4 to 55.2 CP in both the ‘Lapins’ and ‘Brooks’ cultivars. Similar to the results reported by Pope et al. (2015), our chill estimations (mean of 52.3 CP) associated with high yields differed from earlier estimates of the chill that was necessary to overcome dormancy. In our analysis, the optimal chill interval for adequate yields was higher than the CR estimated for budbreak (37 CP for ‘Brooks’ according to Erez (2000) and 35 CP for ‘Lapins’ according to Alburquerque et al. (2008)). Lower CR for budbreak compared to optimal chill intervals for high yield may result from the common practice of using 50% of buds open as a threshold to define the moment of dormancy release in CR assessments. Using such a threshold to estimate CR for budbreak may underestimate the CR needed to reach satisfactory potential productivity in sweet cherry orchards (Campoy et al., 2019).

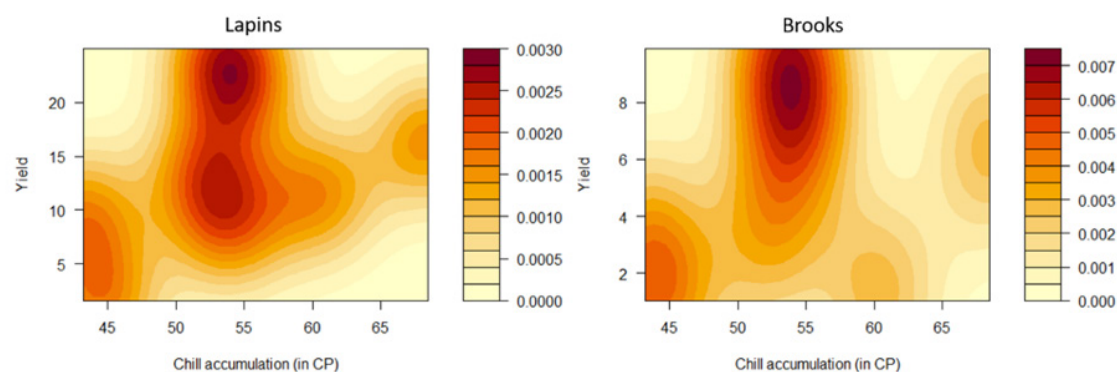


Figure 2. Two-dimensional kernel smoothing between chill accumulation (in CP) and yield (in t ha⁻¹) produced with the ‘chillkernel’ function for two sweet cherry cultivars (‘Lapins’ and ‘Brooks’). Darker colors indicate a greater estimated density (‘z’).

Assuming chill levels that are likely to materialize in the future in this region of Chile, e.g. the expectation of 50 CP by 2050 (likely under the RCP4.5 Representative Concentration Pathway scenario; Fernandez et al. (2020)), we projected probable yield for both cherry cultivars. We based this analysis on a slice of the probability density from the previous kernel density analysis (shown in Figure 2). Given 50 CP, the most likely yield for ‘Lapins’ ranges between 7 and 17 t ha⁻¹. For ‘Brooks’, probable yield may fluctuate between 3 and 6 t ha⁻¹ (Figure 3).

While using only a single point estimate of future chill level (i.e. 50 CP) for projecting future yields might seem unrealistic, utilizing a likely range can adequately represent the inherent uncertainty in climate-related projections. In this regard, we employed the projected range for chill accumulation reported by Fernandez et al. (2020) by 2050 in the RCP4.5 scenario for Quillota for estimating probable yields for both sweet cherry cultivars (Figure 4). This range was between 38 and 63 CP and, according to the yield distribution surface across all chill levels (Figure 2), resulted in the prediction of yields between 7 to 15 t ha⁻¹ for ‘Lapins’

and between 3 and 7 t ha⁻¹ for the cultivar ‘Brooks’. With these ranges for yield, especially values smaller than 8 t ha⁻¹, farmers might experience negative economic results at the end of the season. The latter is especially relevant for the cultivar ‘Brooks’ for which historic yields have been consistently lower than its reported potential yield (8 to 10 t ha⁻¹ according to Programa de Difusión Tecnológica en Cerezos, 2021), probably because of adverse climatic conditions during the growing season. In this scenario, farmers should re-assess the economic viability of growing this cultivar.

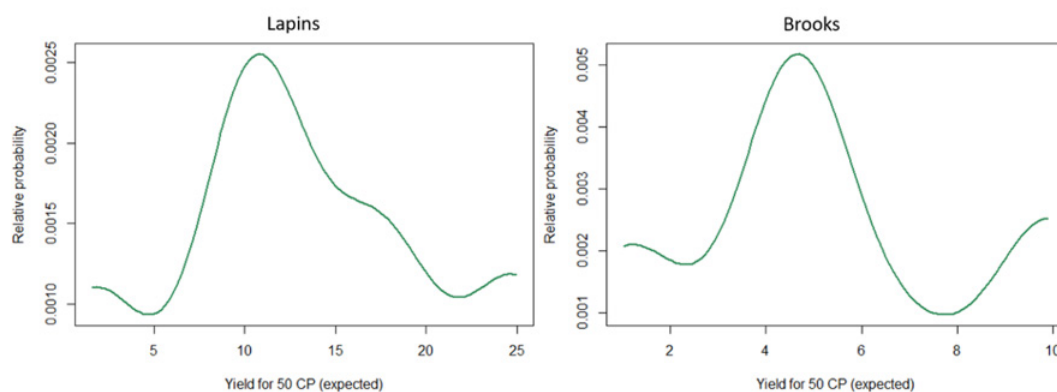


Figure 3. Relative probability of yield (x-axis in t ha⁻¹) based on a projected winter chill accumulation of 50 chill portions for the sweet cherry cultivars ‘Lapins’ and ‘Brooks’ estimated with the ‘chillkernelslice’ function. These yield profiles are extracted from the full distributions of chill-dependent cherry yield shown in Figure 2, considering all data corresponding to chill accumulation of 50 CP.

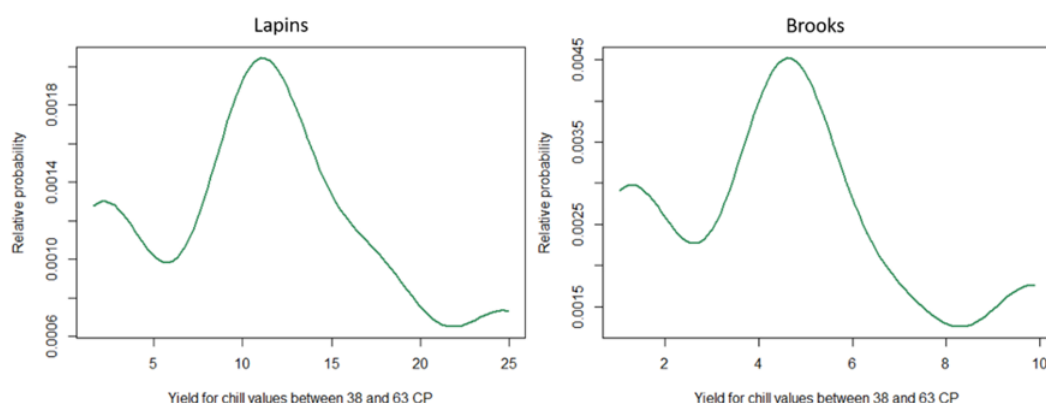


Figure 4. Relative probability of yield (x-axis in t ha⁻¹) in two sweet cherry cultivars given a range of chill accumulation in chill portions (i.e. 38 to 63 CP) estimated with the ‘chillkernelslicerange’ function for future scenarios in Quillota, Chile (Fernandez et al., 2020). These yield profiles are extracted from the full distributions of chill-dependent cherry yield shown in Figure 2, considering all yield predictions for chill accumulation between 38 and 63 CP.

CONCLUSIONS

Here we have demonstrated useful and innovative procedures that generate probabilistic assessments of the relationship between chill accumulation and yield. These can be used to fill in critical missing information and to project probable yields of temperate fruit trees given future chill estimations. The outputs of these procedures can help in managing orchards under climate change, although the methods require further validation and

development. Long-term yield and chill observations may serve to enhance this procedure and produce more reliable outputs. Given issues related to data limitations and access, acknowledging and including factors affecting both chill accumulation and harvest periods is widely expected to improve the analysis and related projections. Generating tools such as the one demonstrated here may help farmers and decision makers in adopting strategies to mitigate the impacts of climate change on temperate fruit tree orchards.

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